

Strengthening Business Continuity Through Cyber Security & Cyber Resiliency

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Cost of a Data Breach report* – Advocating AI & automation

56%

Increase in AI-generated attacks

Attackers are increasingly weaponizing AI, using it to automate the attack lifecycle, scale operations and refine the sophistication of their attacks. This year’s research shows the impact: more than one in four organizations that experienced a malicious attack reported it was AI-driven, a 56% increase over last year. AI deepfake impersonation and AI-enabled malware drove the highest volume of those incidents. The financial consequences: AI-driven attacks added an average of USD 1 million per breach.

41%

Share of ransomware attacks that included threats to brand reputation

Ransomware incidents among breached organizations rose slightly this year over last year (39% versus 34%), as attackers used AI to generate malware and expanded their methods of extortion. While disrupting operations through encrypting remains a key tactic (23%), attackers are shifting to higher-impact pressure methods. Threatening brand reputation is now the most common tactic, cited by 41% of the breached organizations that experienced a ransomware incident. This shift reflects a move away from purely technical disruption toward multilayered extortion strategies that target trust, public perception and long-term business impact.

85%

Share of organizations that plan to increase security spending in response to frontier AI model threats

Originally, 64% of organizations said they would increase security spending after experiencing a breach. Upon learning of new threats from frontier AI models, that figure increased dramatically to 85%. The new threats have also caused organizations to rethink the use of AI agents in security. Three quarters said they will deploy agents at higher rates in alert triage, vulnerability management, and scans and penetration testing.

Ransomware can have devastating downstream effects, negatively impacting an organization’s operations and disrupting people’s ability to access personal finances, healthcare and energy.

Thanks in part to ransomware as a service, these attacks continue to increase. But our research found attackers are shifting their tactics. In addition to encrypting data and demanding a ransom to unlock it, they’re weaponizing brand reputation, increasingly resorting to publicly shaming victims to extort them.

Figure 23. Percentage of ransomware attacks among breached organizations

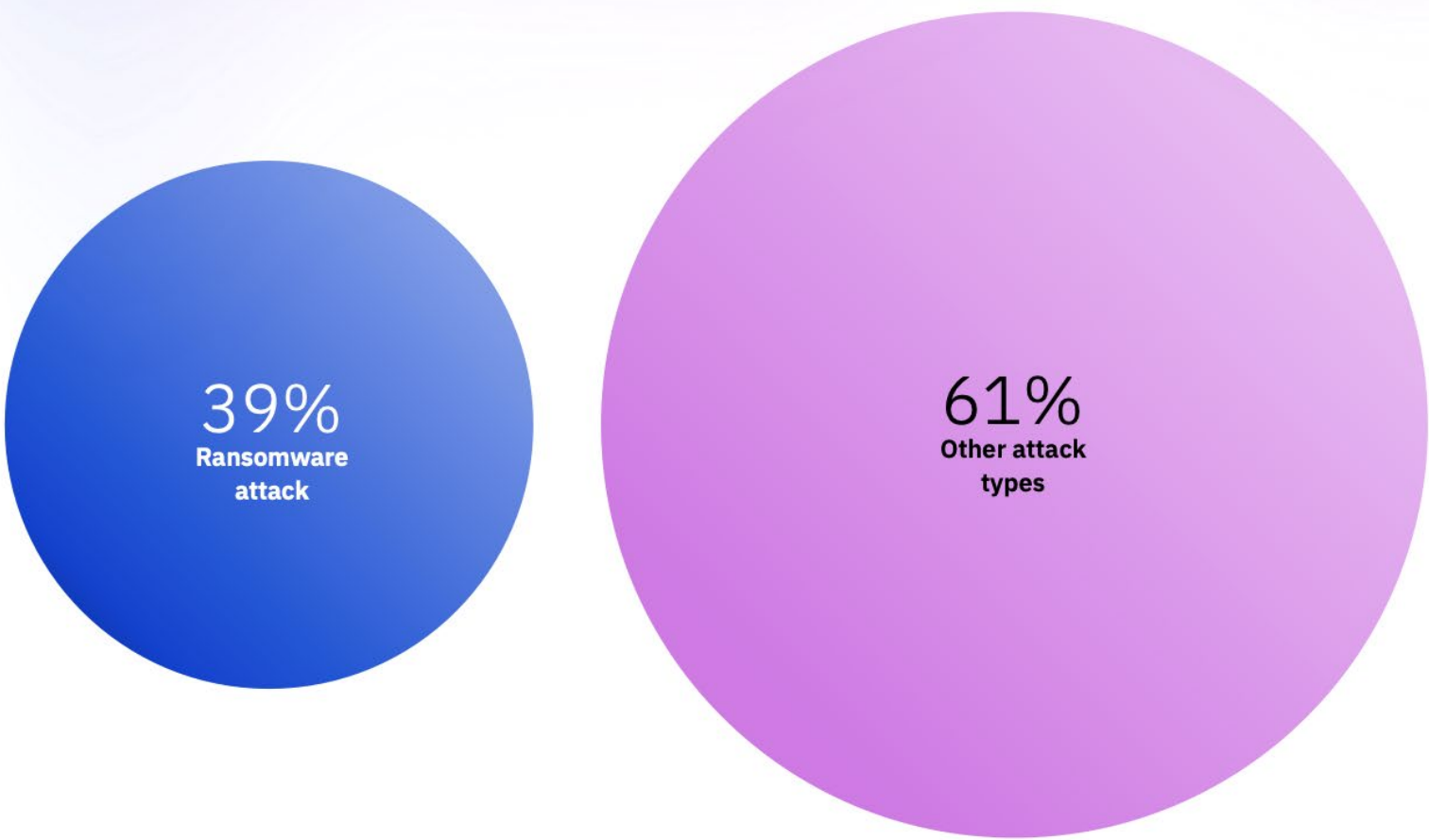
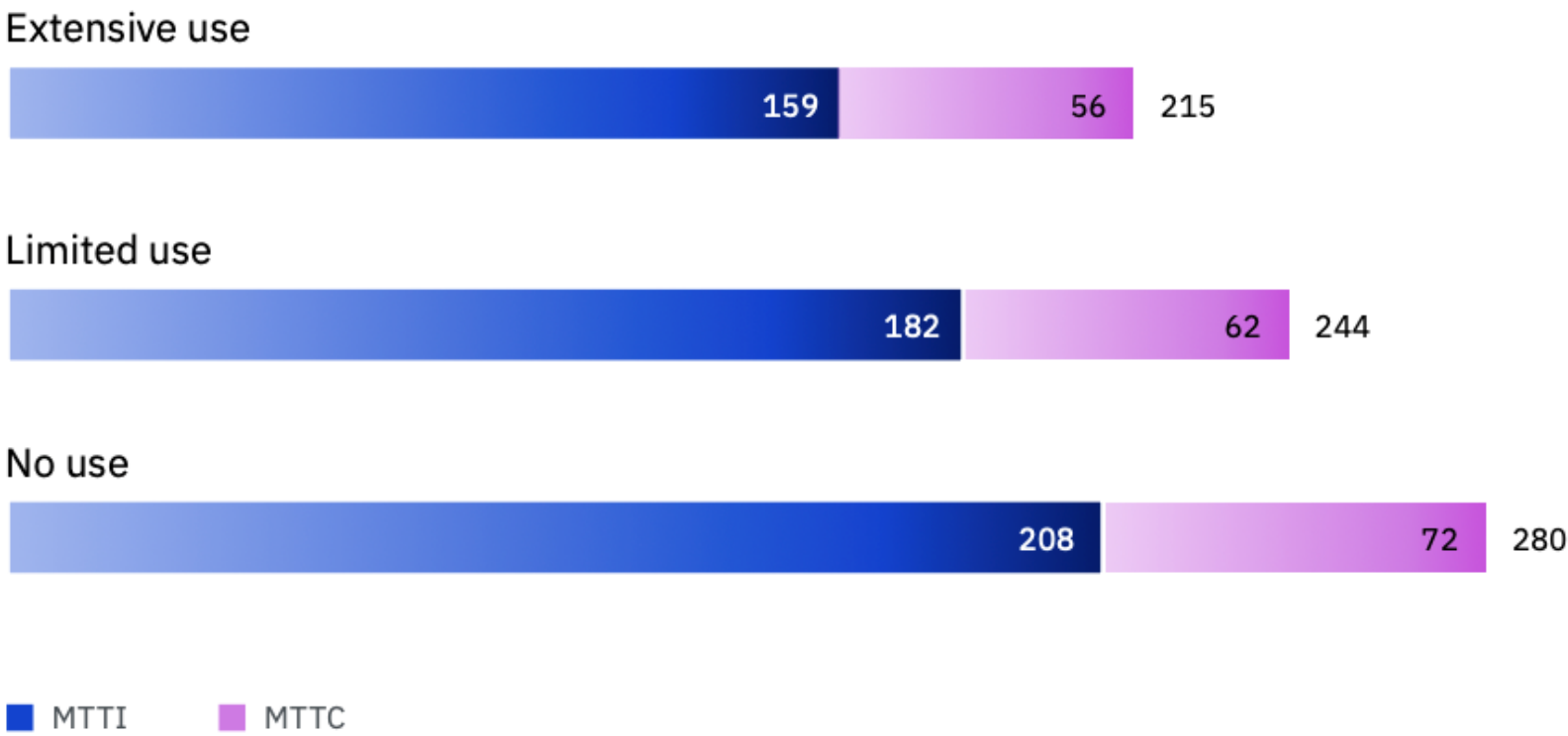


Figure 35. Time to identify and contain a breach with and without AI and automation; measured in days

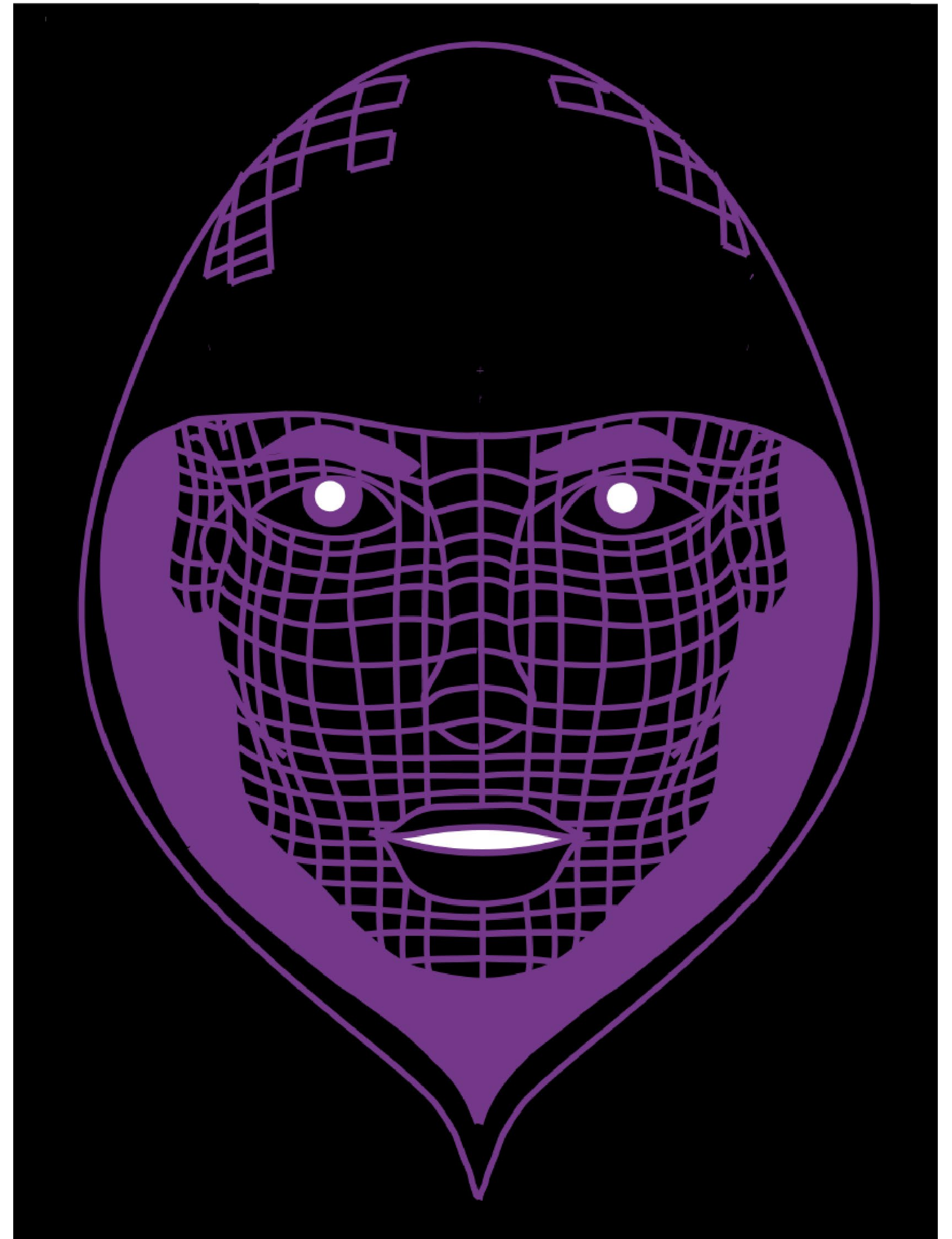


What does a hacker look like?

How do you detect them?

How do you contain them?

How do you recover?



Zero Trust

Use Case:

Insider Threat

Jon Smith is an IT professional.

Jon has access to a lot of sensitive resources.
He ordinarily looks like any other IT professional because he is one.

Jon has encountered a personal financial crisis.
Jon quietly **exfiltrates a massive data breach** to sell on the Dark Web.
He plans to then facilitate a **ransomware attack** for a greater payout.

How quickly do you discover Jon?

How quickly can you contain Jon's activity once you do?

How quickly can you recover to minimize the damage?

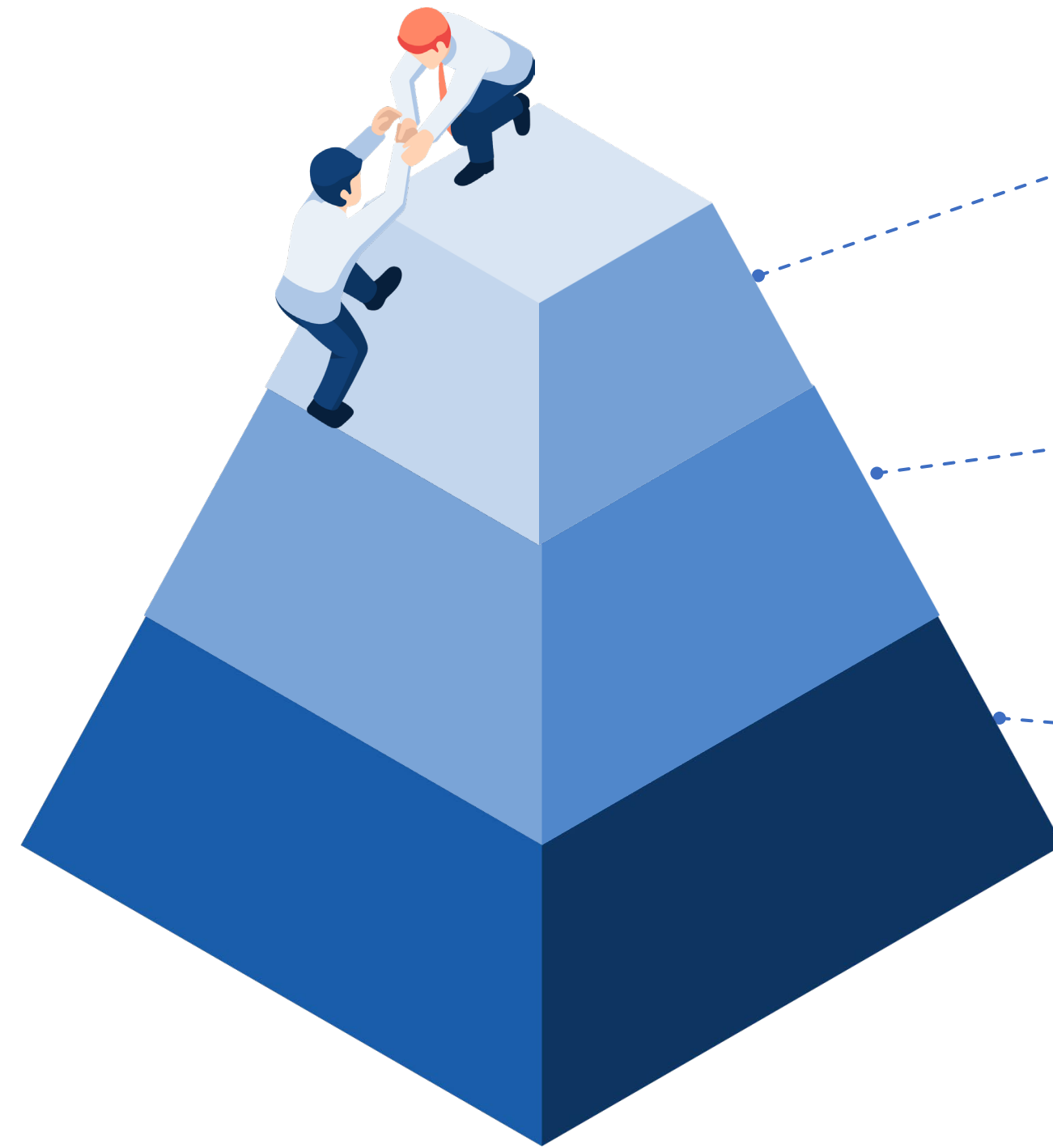
The Cybersecurity Framework context



People, Process, & Technology

Areas of potential risk to assess:

- People: Slow human response, lack of insights, vigilance
- Process: Unimplemented, untested, incomplete
- Technology: Detection and response gaps in traditional security



Solutions

People

Time to respond – automation combined with alerts
Instream recovery guidance – planning ahead
Recover system & data – managing scenarios

Process:

NIST CSF V2, PCI-DSS, DORA standards & regulations
In-house playbooks & other processes

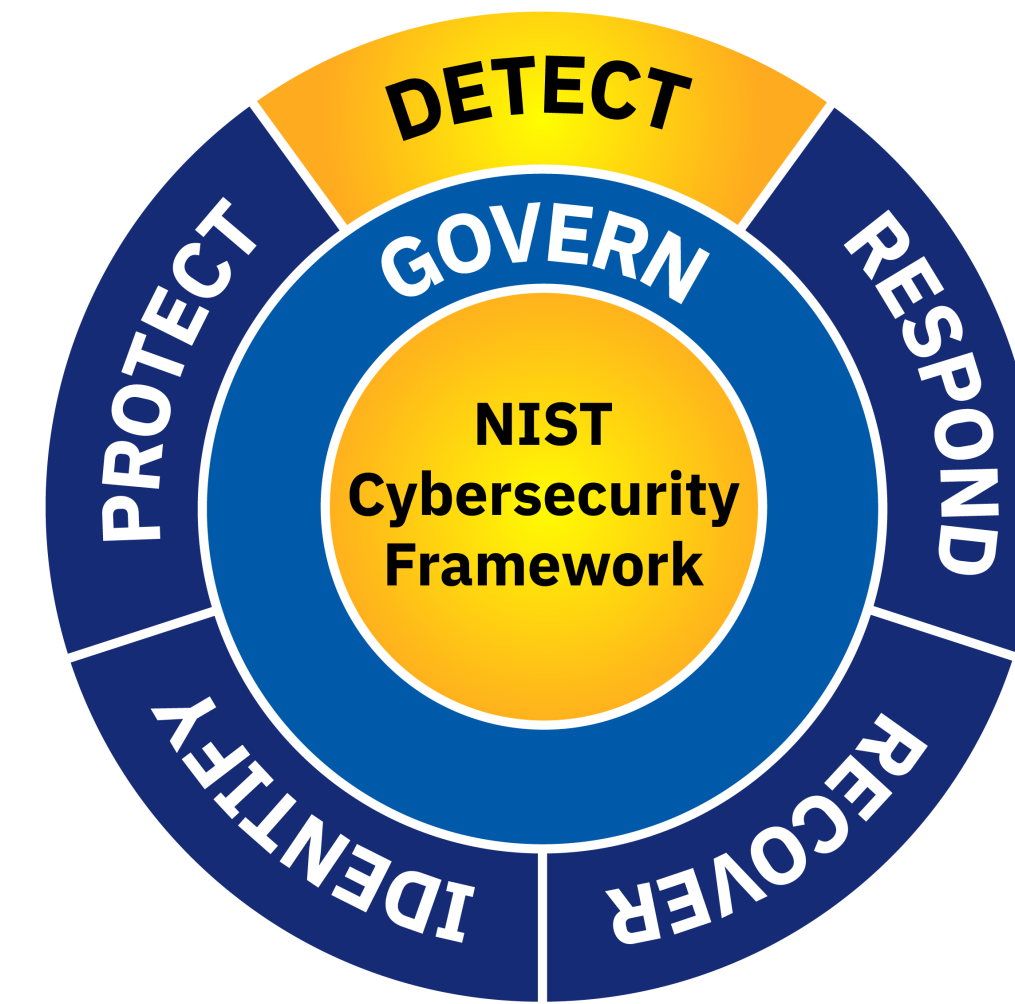
Technology:

Ransomware, exfiltration, & overall behavior detections
Near real-time response to perceived threats
Damage control – containment & recovery

NIST & PCI 4.0 redefine anti-malware

Anti-malware beyond anti-virus...

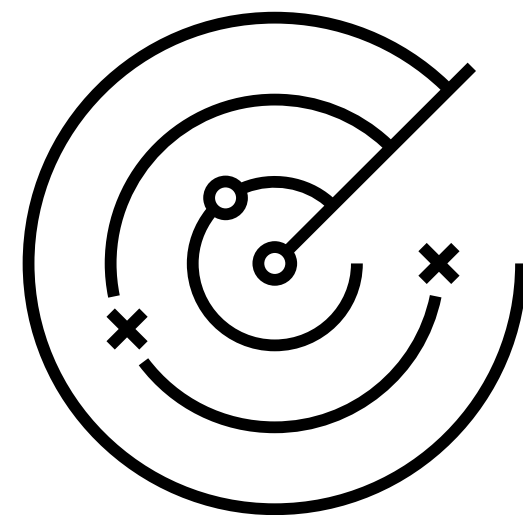
- Non-signature-based **detection**
- **Containment**



Transforming threat management for z/OS

IBM zSecure Detection

- IBM zSecure Networks
- IBM zSecure Threat Monitor
- IBM zSecure Threat Intelligence
- IBM zSecure Alert
- IBM zSecure Adapters for SIEM
- Agentic AI for Detection*



zSecure Networks

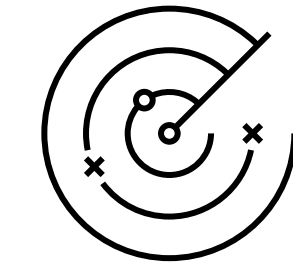
zSecure Networks provides z/OS functionality designed to discover, visualize, and control z/OS Communication Server's TCP/IP network traffic moving in and out of IBM z/OS systems.

Key Features and Capabilities

- **Network Discovery:** Illuminates who and what is accessing the z/OS Communications network traffic, compiling an accurate knowledge base of active connections and their encryption strength
- **Traffic Control:** Validates that established network controls and security policies (such as z/OS and RACF profiles) are correctly configured and operating effectively.
- **Threat Mitigation:** Dynamically creates micro-segmentation for z/OS Communications Server. Restricts users from moving laterally across the network infrastructure with more granular access controls

zSecure Detection

- **zSecure Networks**
- zSecure Threat Monitor
- zSecure Threat Intelligence
- zSecure Alert
- zSecure Adapters for SIEM
- Agentic AI for Detection*



zSecure Threat Monitor

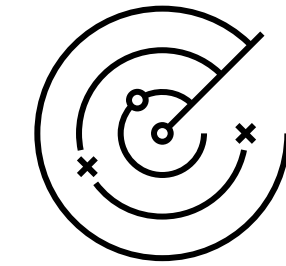
zSecure Threat Monitor provides z/OS functionality designed to monitor for various forms of cyber threats and to respond in near real-time.

Key Features and Capabilities

- **Advanced monitoring:** Monitors potential z/OS threats
 - privilege escalation
 - file integrity
 - accelerated cryptography
 - other forms of potential data corruption
- **Near real-time response:** Contains potential z/OS threats
 - Suspends a suspicious job
 - Revokes & forces a RACF userid
 - Supports custom exclusions lists to minimize false positives

zSecure Detection

- zSecure Networks
- **zSecure Threat Monitor**
- zSecure Threat Intelligence
- zSecure Alert
- zSecure Adapters for SIEM
- Agentic AI for Detection*



zSecure Threat Intelligence

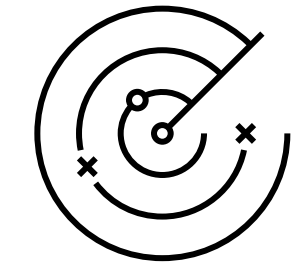
zSecure Threat Intelligence is the refactored and rebranded solution previously named IBM Threat Detection for z/OS.

Key Features and Capabilities

- **Custom AI/ML:** Detects anomalous, potentially malicious, data access on z/OS production systems
- **Leverages high fidelity feeds:** Utilizes a custom feed at the DFSMS level with entitled IBM z/OS Workload Interaction Correlator (zWIC) technology
- **Anti-malware functionality for z/OS:** When combined with containment capabilities such as RACF's userid containment, zSecure Threat Intelligence is designed to help clients in claiming anti-malware based on PCI-DSS 4.0.

zSecure Detection

- zSecure Networks
- zSecure Threat Monitor
- **zSecure Threat Intelligence**
- zSecure Alert
- zSecure Adapters for SIEM
- Agentic AI for Detection*



zSecure Alert & zSecure Adapters for SIEM

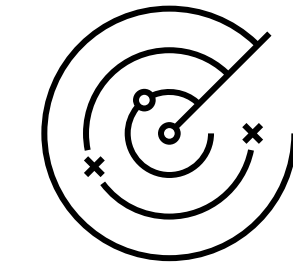
zSecure Alert and zSecure Adapters for SIEM provide rules-based security intelligence custom to z/OS with real-time notification

Key Features and Capabilities

- **z/OS security events:** Detects security events from IBM® z/OS®, RACF®, Db2®, CA, ACF2, CICS®, IMS, Communications Server, Tivoli® Workload Scheduler, Health Checker, and UNIX.
- **Near real-time notification:** Alerts security focal in near real-time methods such as email, text message, and z/OS console message.
- **SIEM integration:** Propagates z/OS security events in real-time to the enterprise level, supporting Security Information and Event Management (SIEM) integration.

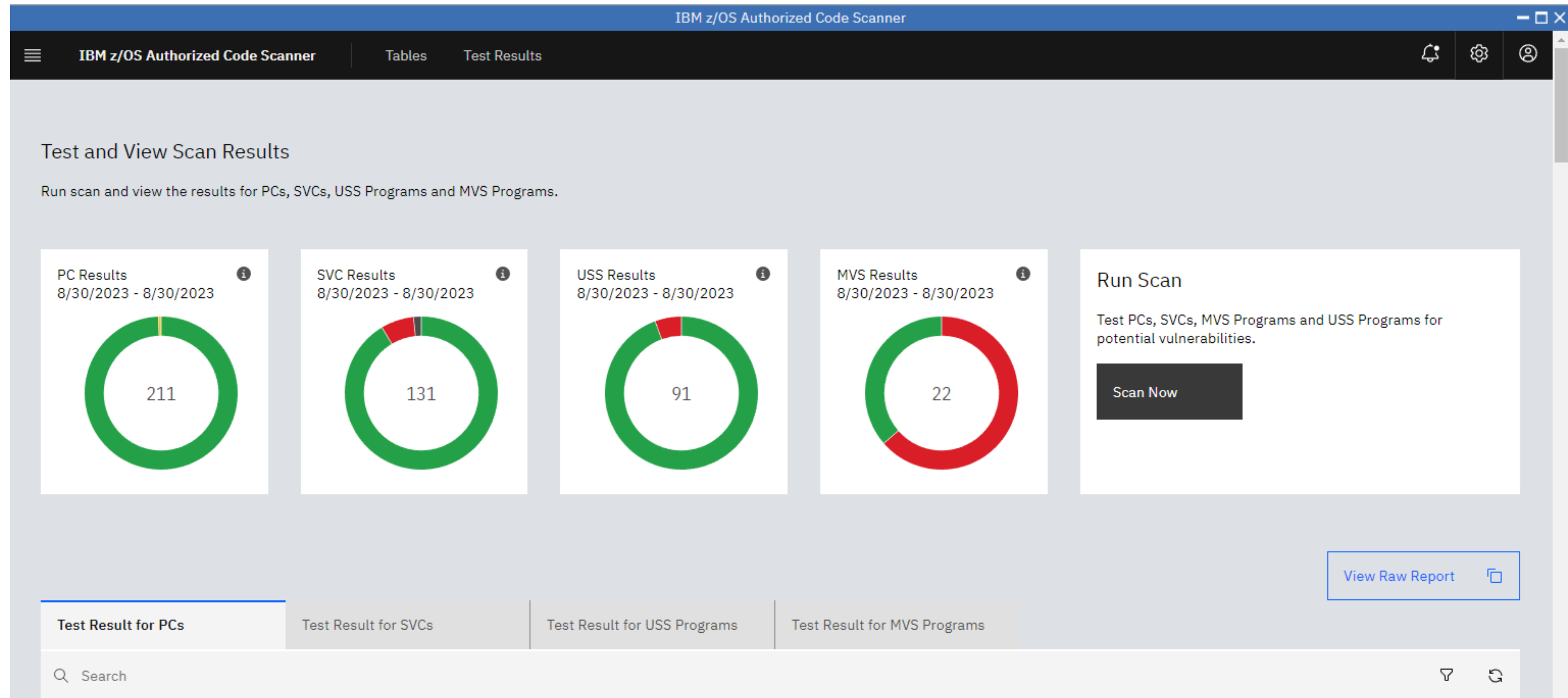
zSecure Detection

- zSecure Networks
- zSecure Threat Monitor
- zSecure Threat Intelligence
- **zSecure Alert**
- **zSecure Adapters for SIEM**
- Agentic AI for Detection*



IBM z/OS Authorized Code Scanner entitlement

An integrity flaw in the z/OS Authorized Program Facility is a **privilege escalation** attack vector to all z/OS userids.



zACS advocacy

"SECINT APARs are not unique to IBM code. All major ISVs have them from time-to-time. zACS helped us find a decades-old exposure lurking in one of our authorized started tasks. The mitigation was trivial, customers were notified, and the exposure was closed. zACS helped Phoenix Software, our customers, and the industry as a whole maintain z/OS's reputation as the most secure and reliable operating system on the planet."

- Ed Jaffe, Chief Technology Officer, Phoenix Software International

Statements of Direction: Networks

IBM intends to add to zSecure Detection*:

- Network data exfiltration monitoring & containment
- Network anomaly detection AI/ML
- Agentic AI support to evaluate z/OS networks

Integration with Cyber Vault

zSecure Detection integrations:

- GDPS/LCP policy-based snapshots from detected anomalies
- Via IZBR, added diagnosis for suspicious userid & cascade of affected data



Transforming threat management for z/OS

Threat visibility

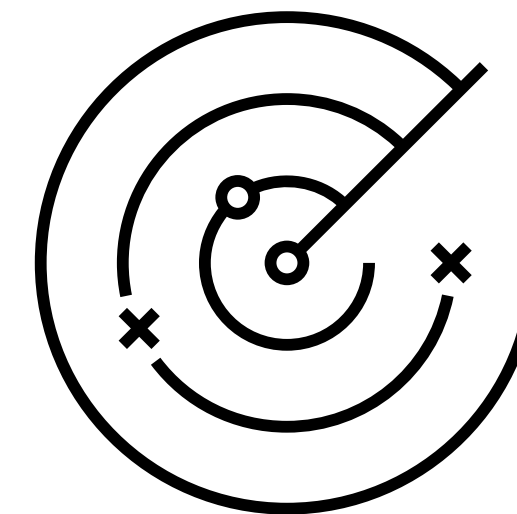
- Networks
- Cryptography
- Access patterns
- Privilege escalation
- File and system integrity
- Overall system activity

Near real-time response & recovery

- Security focal notification
- SIEM propagation
- Suspicious program suspension
- Userid revocation
- Cyber Vault snapshot integration
- Cyber Vault recovery integration

zSecure Detection

- zSecure Networks
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Thank you

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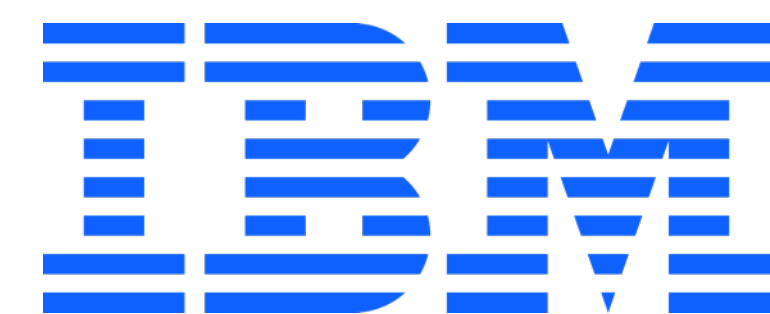
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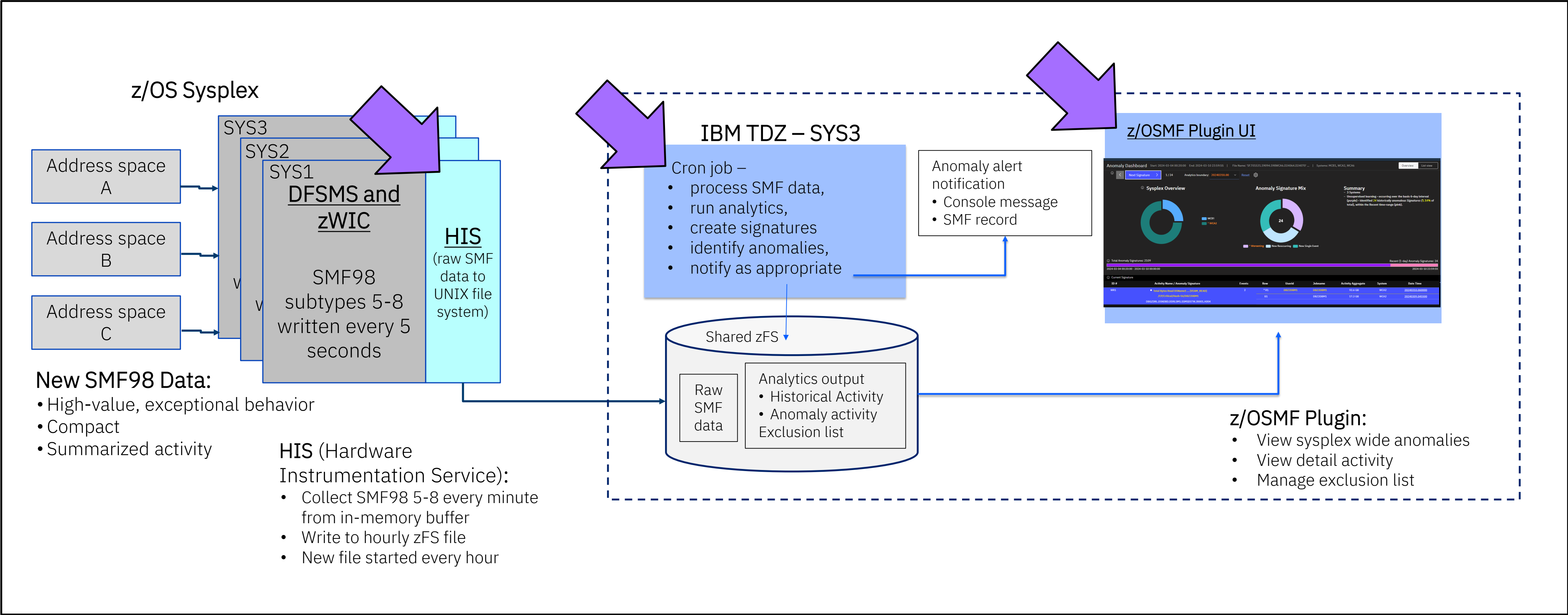
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Backup

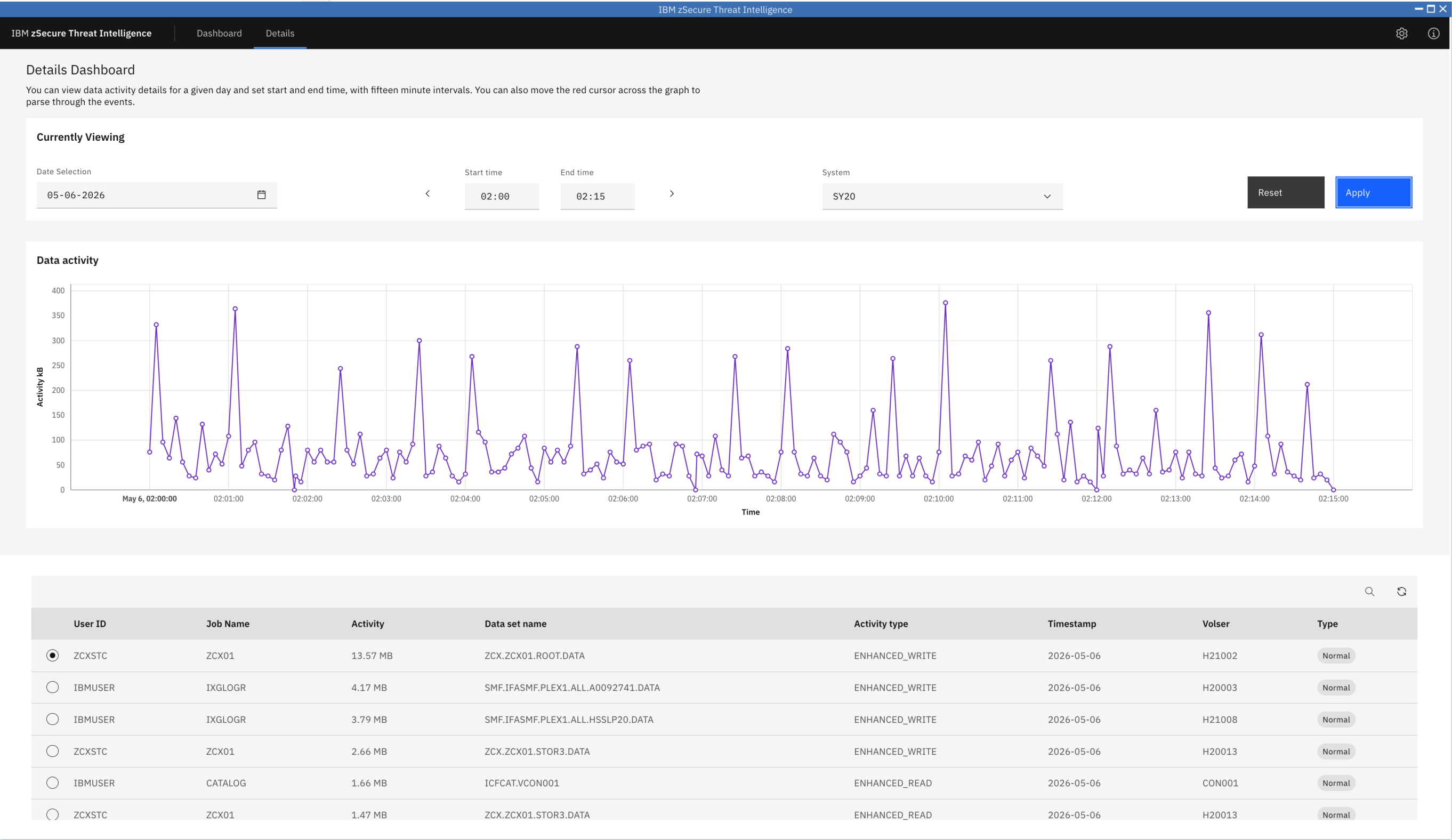
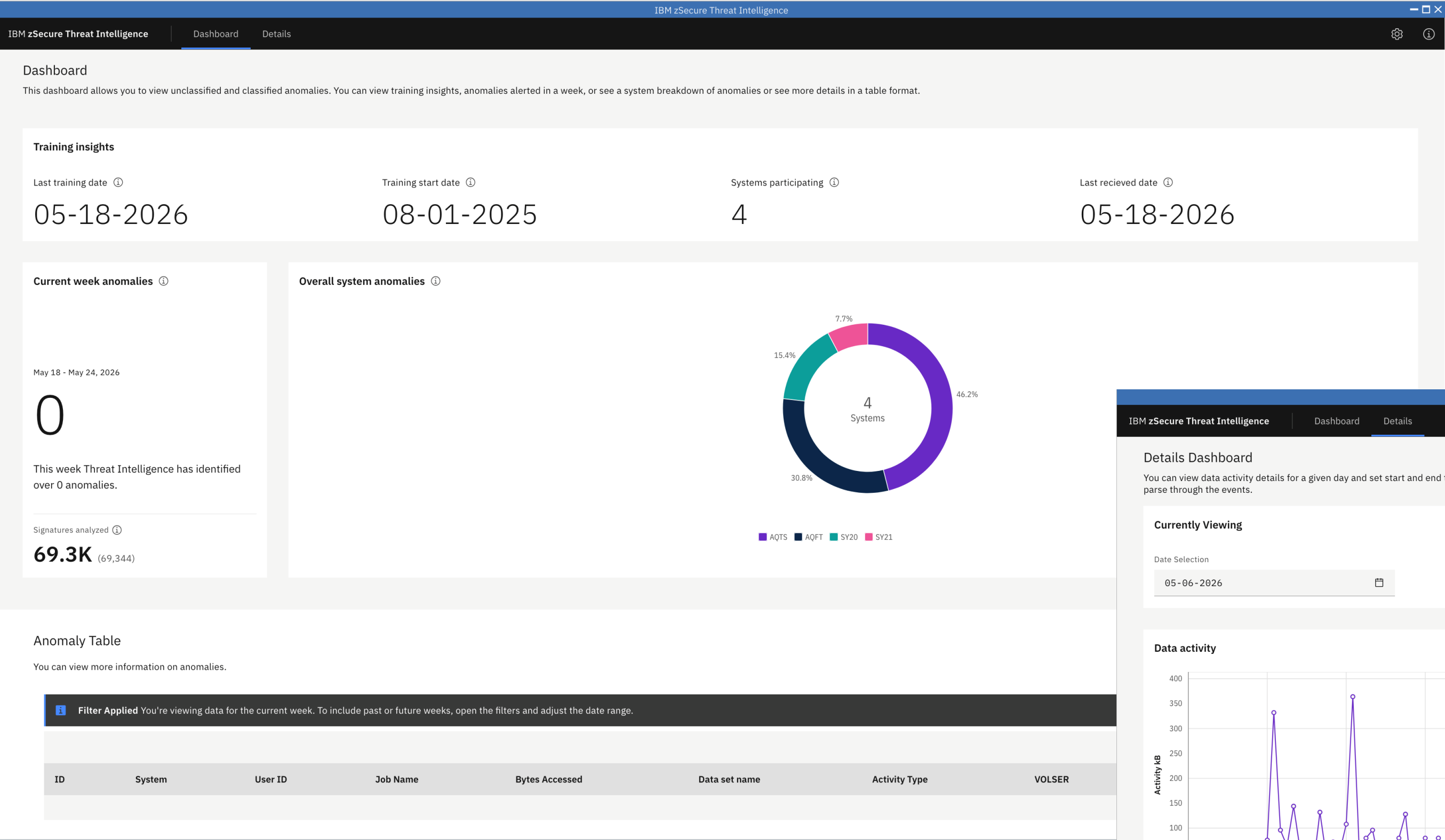
Architectural shift for zSecure Threat Intelligence



The zSecure Threat Intelligence engine

- ➡ **Common Data Provider (CDP)** feed for SMF
 - Store and then feed **months of SMF data**
 - Remove SMF logstream requirement
- ➡ **Machine Learning**
 - Greater **explainability** from industry-standard AI/ML libraries
 - Flexibility in refining & **customizing** the solution
- ➡ **New UX**
 - Reflect **custom insights** from the Machine Learning solutions
 - **Carbon-compliant** UX simplifies usage

zSecure Threat Intelligence refactored UX



RACF userid containment

The containment aspect of z/OS anti-malware

Extending RACF userid revocation

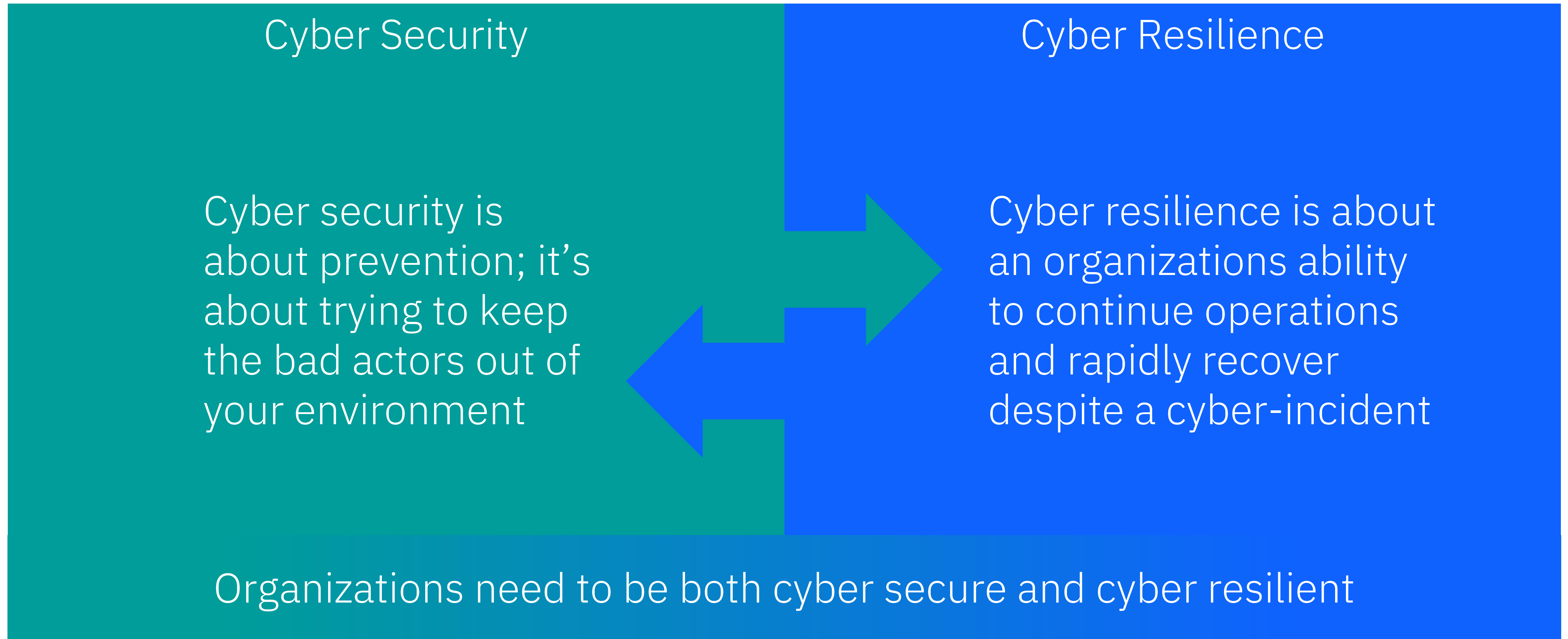
- Subsequent RACF **access checks fail**
- Sysplex-wide **notification** signal
- **Revocation still occurs** on next logon

Providing safeguards

- Protected with a new custom **RACF profile**
- Supports **containment reversal** as needed
- **Exclusion lists** support proactive planning



zSecure Detection: bridging Cyber Security to Cyber Resilience



Accelerating adoption of IBM Z Cyber Vault automation

202

LCP Manager licenses now
in use worldwide

72

Unique clients have LCP
Manager deployments

32%

Of the worlds top 25 banks
are using LCP Manager

Protect your data with IBM Safeguarded Copy

- zSecure Detection integrations with Cyber Vault:
 - GDPS/LCP policy-based snapshots from detected anomalies
 - Via IZBR, added diagnosis for suspicious userid & cascade of affected data

- Prevent sensitive point in time copies of data from being modified or deleted due to errors, malicious destruction or ransomware attacks.
- Create up to 1,024 SafeGuarded Backups for a production volume stored in SafeGuarded Backup Capacity, which is not accessible to any server.
- The data is accessible only after a SafeGuarded Backup is recovered to a separate recovery volume.
- IBM GDPS or IBM CSM required to create and manage Safeguarded Backups

